



Chapter 38- Wireless LAN Controller (WLC) & Access Point (AP)

◆ 1. What is AP (Access Point)?

👉 An **Access Point (AP)** is a device that provides **Wi-Fi signal** so devices like laptops, phones, PCs can connect to the network.



Simple Understanding:

- AP = Wi-Fi router (in simple terms)
- It only **sends and receives wireless signals**



Key Points:

- Each AP has:
 - **SSID (Name)** → Wi-Fi name you see
 - **Password** → For security
 - **BSSID** → Unique MAC address of each AP
- If there are **120 APs**, each one must be configured separately → ❌ very difficult



Daily Life Example:

Think of AP like **fans in different rooms**

- Each fan works individually
 - You must turn each fan ON/OFF manually
-

◆ 2. What is WLC (Wireless LAN Controller)?

👉 WLC is a **central device** that manages multiple APs.

🧠 Simple Understanding:

- Instead of configuring 120 APs separately → use **1 WLC**
- WLC will **push configuration (SSID, password)** to all APs automatically

📌 Key Points:

- APs connect to **switches**
- WLC sends configuration to all APs
- Easy management 👍

🏠 Daily Life Example:

Think of WLC like a **main electricity control panel**

- You control all fans (APs) from one place ⚡

◆ 3. WLC Capacity (Important for Exams ⚠)

Type	Model Series	AP Support
🟢 Small	2504	~75 APs
🟡 Medium	3504	100–200 APs
🟡 Large	5520	1000+ APs
🔴 Enterprise	8540	5000+ APs

◆ 4. How Wi-Fi Works

- PC connects to Wi-Fi
 - Wi-Fi sends **signals only**
 - Devices send **probe requests** to find networks
-

SERVICE SETS (MBSS – Mesh Network)

◆ What is MBSS?

 Used when **Ethernet cable cannot reach every AP**

Key Concepts:

- Creates a **Backhaul Network** (AP to AP communication)
- One AP is connected to wired network → called **RAP (Root AP)**
- Other APs → called **MAPs (Mesh APs)**

Mesh AP Features:

- Uses **2 Radios**:
 - One for clients
 - One for AP-to-AP communication
- Uses protocol to find **best path** (like routing protocols)

Example:

Think of **no internet cable in a village**

- One house has internet (RAP)
 - Others connect wirelessly (MAPs)
-

◆ DISTRIBUTION SYSTEM (DS)

👉 Wireless networks are usually connected to **wired network**

📌 Key Points:

- Wired network = **Distribution System (DS)**
- Each Wi-Fi network (SSID) → mapped to a **VLAN**
- AP can provide **multiple SSIDs**
- Each SSID:
 - Has unique **BSSID**
 - Connected via **TRUNK port**

🏢 Example:

Office Wi-Fi:

- Staff Wi-Fi → VLAN 10
- Guest Wi-Fi → VLAN 20

◆ ADDITIONAL AP MODES

🔄 1. Repeater Mode

- Extends Wi-Fi range
- Re-transmits signals

📉 Problem:

- Same channel → reduces speed
-

2. Workgroup Bridge (WGB)

- Connects **wired devices to Wi-Fi**

Example:

- PC has no Wi-Fi
 - Connect PC → WGB → AP
-

3. Outdoor Bridge

- Connects networks over long distance
- Uses special antennas

Types:

- Point-to-Point
 - Point-to-Multipoint
-

◆ TYPES OF AP (Architecture)

◆ Lightweight AP (Important ★)

Concept: Split-MAC Architecture

 Work is divided between AP and WLC

◆ AP Functions (Real-time ⌚)

- Sending/receiving signals
 - Encryption/Decryption
 - Beacon/Probe sending
 - Packet priority
-

◆ WLC Functions (Non real-time 🧠)

- RF Management
 - Security & QoS
 - Authentication
 - Roaming
 - Resource allocation
-

🔒 Security:

- AP & WLC authenticate using **X.509 Certificates**
-

◆ Protocol Used:

👉 CAPWAP (Control and Provisioning of Wireless AP)

- Based on LWAPP

🔍 Two Tunnels:

1. **Control Tunnel (UDP 5246) 🔒**
 - Configuration & management
 - Encrypted
 2. **Data Tunnel (UDP 5247) 📦**
 - Client traffic goes to WLC
 - Not encrypted by default
-

🔌 Important:

- AP connects to **Access Port (not trunk)**
 - All traffic goes to WLC
-

⚠ Important Concept:

- If WLC goes DOWN → All APs go DOWN ❌ (in Lightweight mode)
-

◆ Traffic Flow:

PC → AP → Switch → WLC

👉 WLC **does NOT read data**, just manages it

◆ LIGHTWEIGHT AP MODES

● 1. Local Mode (Default)

- Provides Wi-Fi to clients
-

● 2. FlexConnect Mode

- Works like Local Mode
 - If WLC goes DOWN:
 - AP continues working using stored config ✅
-

🔍 3. Sniffer Mode

- Captures Wi-Fi packets
- Used with tools like Wireshark

👉 Used to check:

- Who is browsing what
 - Deep packet inspection
-

4. Monitor Mode

- Detects rogue devices
 - Sends **de-authentication messages**
-

5. Rogue Detector

- Does NOT use radio
 - Checks wired network traffic
 - Uses ARP + WLC data
-

6. Bridge / Mesh Mode

- Connects sites wirelessly
-

7. Flex + Bridge Mode

- Mesh + FlexConnect
 - Works even if WLC fails
-

Quick Summary:

- Sniffer → Capture packets
 - Monitor → Detect hackers
 - Rogue Detector → Check wired network
 - Bridge → Connect sites
 - FlexConnect → Works without WLC
-

CLOUD-BASED APs

 Example: Cisco Meraki

Features:

- Managed via **cloud dashboard**
- Central configuration
- Reports & monitoring

Important:

- Data traffic → goes to wired network
 - Only control traffic → goes to cloud
-

◆ WLC Deployment Types

1. Unified WLC

- Hardware device
 - Supports ~6000 APs
-

2. Cloud-based WLC

- Runs as VM
 - Supports ~3000 APs
-

3. Embedded WLC

- Inside switch
 - Supports ~200 APs
-

4. Mobility Express

- WLC inside AP

- Supports ~100 APs
-



Resume Building Tips



Step 1:

- Use tools like:
 - ChatGPT
 - Canva
-



Step 2:

- Logout before using ChatGPT
 - 👉 To avoid old data influence
-



Step 3:

- Choose template in Canva
-



Step 4:

Use ChatGPT prompts:

- “Write professional summary for network engineer with 1 year experience”
 - Trim content if too long
-

Step 5:

Sections to include:

- Summary
 - Skills
 - Experience
 - Education
-

Education:

- Add 12th + Graduation
 - Add CGPA only if good
-

Skills:

- Ask:
 “Provide technical skills for network engineer resume”
-

Experience:

Example:

- Worked at **Humming Byte Technology**
 - Add client project
-

Interview Preparation:

- Ask ChatGPT:
 “Provide story for 1 year experience as network engineer”
-

 Tip:

- Practice in local language first
 - Then speak in English
-

 Final Summary

- ✓ AP = Provides Wi-Fi
 - ✓ WLC = Controls multiple APs
 - ✓ BSSID = MAC of AP
 - ✓ CAPWAP = AP ↔ WLC communication
 - ✓ Lightweight AP = Needs WLC
 - ✓ FlexConnect = Works even if WLC down
 - ✓ Sniffer = Capture packets
 - ✓ Monitor = Detect rogue devices
 - ✓ Bridge = Connect networks
-

 Remember:

- **CAPWAP ports: 5246 & 5247**
 - **WLC manages APs centrally**
 - **FlexConnect = Backup mode**
-



SUMMARY



Access Point (AP)

- AP provides Wi-Fi to devices
 - Each AP has:
 - SSID (Wi-Fi Name)
 - BSSID (MAC Address)
 - Without WLC → manual configuration
-



Wireless LAN Controller (WLC)

- Central device to manage multiple APs
 - Pushes configuration to all APs automatically
 - Reduces manual work
-



CAPWAP Protocol

- Used between AP ↔ WLC
 - Ports:
 - UDP 5246 → Control
 - UDP 5247 → Data
-



Lightweight AP (Important)

- Works with WLC
 - Uses Split-MAC Architecture
-

AP Modes

- Local → Normal Wi-Fi
 - FlexConnect → Works without WLC
 - Sniffer → Packet capture
 - Monitor → Detect attacks
 - Bridge → Connect networks
-

VLAN Mapping

- Each SSID → mapped to VLAN
 - Uses trunk on switch
-

Mesh Network (MBSS)

- RAP → Root AP (wired)
 - MAP → Mesh AP (wireless)
-

Important

- If WLC down → AP down (except FlexConnect)
-

CONCLUSION

 In enterprise networks:

- AP provides Wi-Fi 
- WLC manages everything centrally 

 Final understanding:

AP = Worker 

WLC = Manager 

Q & A

? Q1: What is an Access Point?

✓ Answer:

An Access Point is a device that provides wireless connectivity to devices.

👉 It converts wired network → wireless signals 📶

? Q2: What is WLC?

✓ Answer:

WLC (Wireless LAN Controller) is a central device that manages multiple access points.

👉 It simplifies configuration and management

? Q3: Why WLC is needed?

✓ Answer:

Without WLC:

- Must configure each AP manually ❌

With WLC:

- Centralized management ✓
 - Easy configuration
 - Better security
-

? Q4: What is SSID?

✓ Wi-Fi network name

? Q5: What is BSSID?

✓ MAC address of AP radio

👉 Each AP = unique BSSID

? Q6: What is CAPWAP?

✓ **Answer:**

CAPWAP is a protocol used for communication between AP and WLC.

? Q7: CAPWAP ports?



- UDP 5246 → Control 🔒
 - UDP 5247 → Data 📦
-

? Q8: Difference between Control & Data tunnel?

Tunnel Purpose

Control Configuration

Data Client traffic

? Q9: Is CAPWAP data encrypted?

✓ Control → Encrypted

✗ Data → Not encrypted (by default)

? Q10: What is Split-MAC Architecture?

✓ Answer:

Work is divided between AP and WLC.

👉 AP → real-time tasks

👉 WLC → management tasks

? Q11: Functions of AP?



- Signal transmission 📶
 - Encryption
 - Beacon/probe
-

? Q12: Functions of WLC?



- Security 🔒
 - QoS
 - Roaming
 - RF management
-

? Q13: What is Local Mode?

✓ Default mode → provides Wi-Fi

? Q14: What is FlexConnect Mode?

✓ Works even if WLC goes down

👉 Stores config locally

? Q15: What is Sniffer Mode?

✓ Captures packets for analysis

👉 Used with Wireshark

? Q16: What is Monitor Mode?

✓ Detects rogue devices

? Q17: What is Rogue Detector?

✓ Detects unauthorized devices on wired network

? Q18: What is Bridge Mode?

✓ Connects two networks wirelessly

? Q19: How SSID is mapped?

✓ Each SSID → mapped to VLAN

👉 Example:

- Staff → VLAN 10
 - Guest → VLAN 20
-

? Q20: AP connects to trunk or access port?

✓ Access Port ✓

? Q21: Why trunk used in wireless?

✓ Between switch and network (for VLANs), not AP

? **Q22: What is Mesh Network?**

- ✓ AP-to-AP wireless communication
-

? **Q23: What is RAP?**

- ✓ Root Access Point (wired)
-

? **Q24: What is MAP?**

- ✓ Mesh Access Point (wireless)
-

💡 **SCENARIO QUESTIONS**

? **Q25: If WLC goes down, what happens?**



- Lightweight AP → stops working ✗
 - FlexConnect → continues working ✓
-

? **Q26: Company has 500 APs. What solution?**

- ✓ Use WLC for centralized management
-

? **Q27: User reports Wi-Fi slow. What to check?**



- Channel overlap
- Interference
- AP load
- Signal strength

? Q28: AP not joining WLC. Reasons?



- Network issue
- CAPWAP blocked
- Wrong VLAN
- No IP

? Q29: How AP joins WLC?



Steps:

1. AP gets IP (DHCP)
2. Finds WLC (DNS/DHCP/manual)
3. Establish CAPWAP tunnel

? Q30: How to verify AP status?



Commands:

- show ap summary
- show capwap client

? Q31: How to check CAPWAP?



- Check UDP ports 5246/5247
 - Ping WLC
-

? Q32: Troubleshooting Wi-Fi issue?

✓ Steps:

- Check AP status
- Check WLC
- Check VLAN mapping
- Check RF interference

? Q33: Difference between Lightweight AP & Autonomous AP?

Feature	Lightweight	Autonomous
Control	WLC	Local
Config	Centralized	Manual
Use	Enterprise	Small network

? Q34: What is Mobility Express?

✓ WLC inside AP

? Q35: What is Cloud-based AP?

✓ Managed via cloud dashboard ☁

🎯 🔥 One Liner

“Explain WLC and AP in simple way”

✓ Perfect Answer:

An Access Point provides wireless connectivity to users, while a Wireless LAN Controller centrally manages multiple APs. The AP handles real-time traffic, and the WLC handles configuration, security, and management. They communicate using CAPWAP protocol.